

TremorLens

SPANDAN Engine · Team_alpha · AI Arena 3.0 · AI Vision

The smartphone in your pocket is a structural health instrument. Every bridge, building, fan, and machine vibrates at its own natural rate, set by its stiffness. When a structure weakens, that rate drops. TremorLens reads this vibration two ways: the phone's **camera** measures it from a distance, with nothing touching the structure, and the phone's built-in **motion sensor** confirms it by touch. Point, wait sixty seconds, and the structure tells you how it feels.

Why this matters

India has inventoried 1,72,517 highway structures, and not one has a published record of its vibration signature. Inspection remains visual. The Gambhira bridge collapse of July 2025 was caused by a crushed support that no eye could see; across 2021–25, India recorded roughly 170 bridge collapses and 202 deaths. The Ministry of Road Transport and Highways now mandates inspections before and after every monsoon — a rule with no affordable instrument behind it. A smartphone is that instrument: 85% of Indian households already own one.

How it works, in four steps

1. **Watch.** Any phone camera films the structure. Small printed stickers give the camera high-contrast points to track.
2. **Measure.** Software tracks those points to a fiftieth of a pixel — sub-millimetre motion, invisible to the eye. Camera shake is cancelled using stationary background points.
3. **Listen.** The motion becomes a vibration fingerprint: the structure's natural rate, how quickly its vibrations fade, and how clean or distorted its motion is.
4. **Judge.** The SPANDAN engine compares today's fingerprint with the structure's own healthy record. It says whether something changed, where on the structure, what kind of fault fits best, and how confident it is — and it refuses to answer when the view is not good enough.

What the tests show

Check	Result
Vibration rate, camera vs known ground truth	within 0.31%
Camera vs phone motion sensor, same structure	within 0.10%
Damaged structure vs its healthy twin; blind test of three	flagged correctly; no false alarm on the healthy repeat
167 Hz read by a 60 fps camera (rolling-shutter method)	0.00% error
Every number above	regenerates from one command, shipped in this package

The verdict screen

One phone films, and this is what the engineer sees: a health score out of 100, the healthy and current vibration rates, the change between them, and the camera cross-checked against a phone motion sensor resting on the structure. All headline numbers come from a physics-faithful synthetic test rig that ships with the source code, so any reviewer can reproduce them in one click.

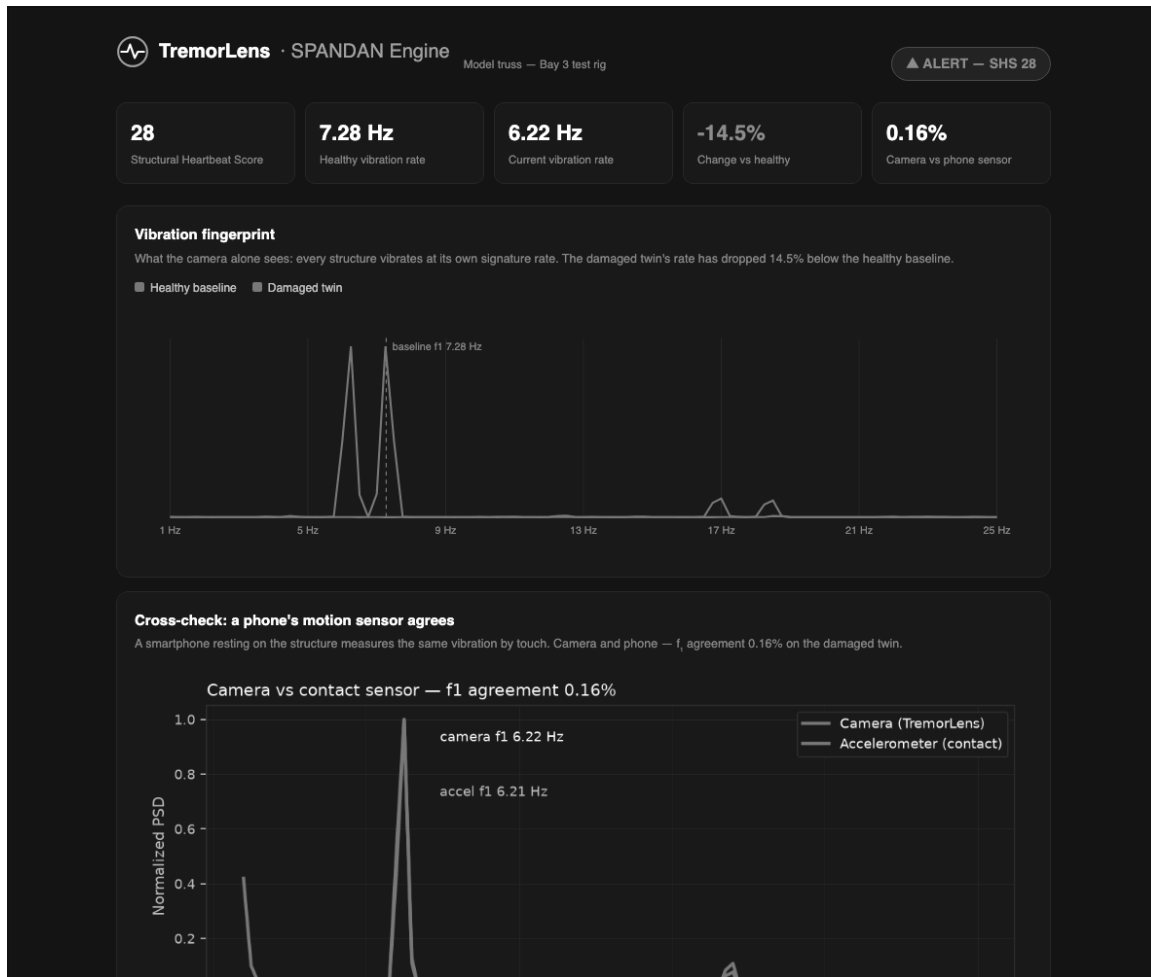


Figure 1 — The verdict dashboard: health score, vibration fingerprint, and the camera–phone cross-check, generated by one command.

Reading the screen

- **The score** starts at 100 for a healthy structure and falls as the vibration rate drifts from the healthy record; the red banner appears only when the drift is too large to be weather or measurement noise.
- **The fingerprint chart** overlays the healthy record and today's reading — when the tall peak moves left, the structure has become less stiff.
- **The cross-check chart** shows the camera and the phone's motion sensor measuring the same vibration two independent ways and agreeing — the reason the number can be trusted.

The fleet view

The same engine scales from one phone to a city's existing cameras. Each structure keeps its own record; the operator sees every verdict in one place.

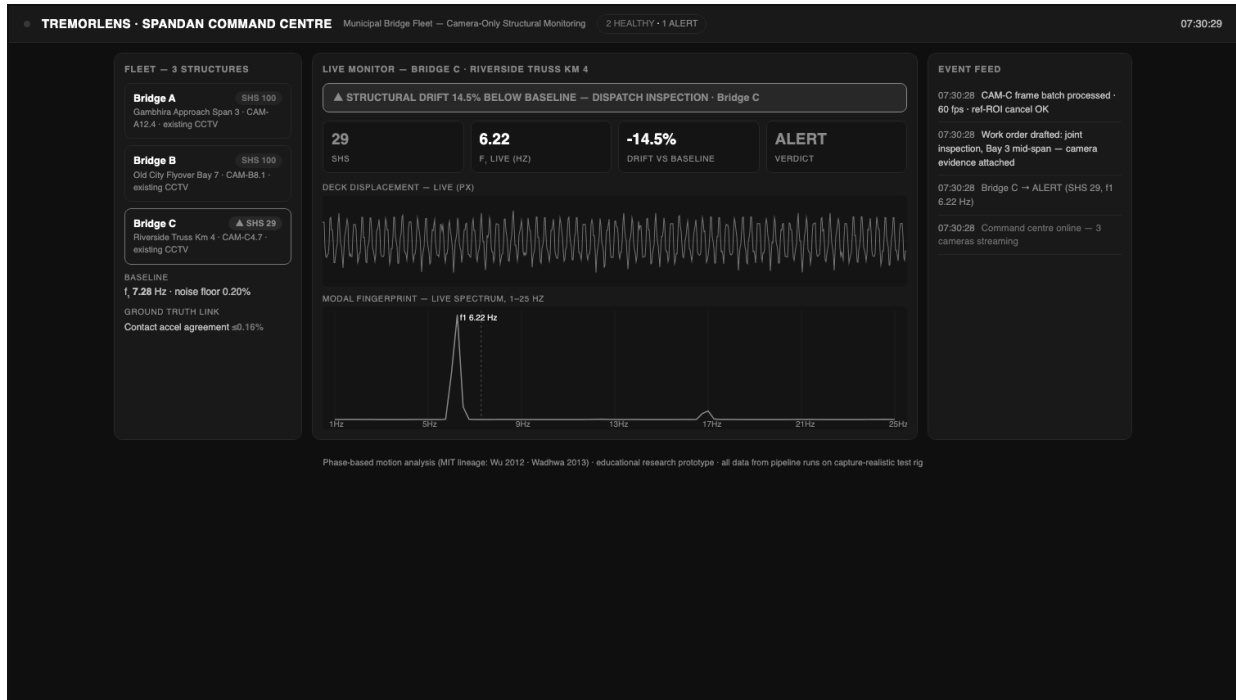


Figure 2 — The command centre: every structure, every verdict, on cameras a city already owns.

Beyond bridges: the household instrument

- **Ceiling fans** (410 million installed): point the phone up, learn whether the wobble is a loose blade or a failing mount.
- **Furniture and school equipment:** a loosening bolt lowers a chair's vibration rate — measurable in ninety seconds, before and after tightening.
- **Farm pumps and appliances:** a motor drifting toward burnout changes its vibration signature weeks before it fails.
- **Homes and rentals:** a sixty-second baseline for a balcony or railing, kept as a record, compared after every monsoon or tremor.

Honesty, by design

TremorLens labels every verdict as screening, never certification. It publishes its error bars, declares a guard band for temperature effects, rejects measurements taken from a shaking hand or a featureless surface, and its fault-naming model answers only after the detection layer has flagged a change — with a stated confidence set, so it can honestly say it is unsure.